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01 Target Distribution

What it shows: This bar chart displays the count and percentage distribution of the binary target variable 'churn', comparing retained customers (Negative/0) against churned customers (Positive/1).

Key finding: Out of 64,374 total records, 47.4% (30,493) have churned, representing an exceptionally high churn rate that is nearly equal to the 52.6% (33,881) who remained.

Business implication: A churn rate near 50% indicates a severe product-market fit or customer experience crisis, requiring the retention team to urgently implement targeted save-campaigns before customer acquisition costs entirely erode profitability.

Confidence caveat: While the large sample size of over 64,000 records provides high statistical power, this high proportion of churn may reflect a balanced, synthetic dataset rather than the actual organic market rate.

Suggestion: Execute a segmented correlation analysis on tenure, contract type (e.g., Month-to-month), and payment method to identify the primary churn drivers and deploy immediate intervention strategies.

02 Rate By Gender

What it shows: This horizontal bar chart compares the customer churn rate (labeled as "Positive Rate") by gender, plotting the percentages for female and male customer segments.

Key finding: Female customers exhibit a significantly higher churn rate of 55.0% compared to male customers at 38.6%, representing a 16.4 percentage point difference.

Business implication: The retention team must prioritize targeted engagement strategies, feedback surveys, and tailored loyalty incentives specifically designed for female customers to address this disproportionate churn risk.

Confidence caveat: This simple correlation does not establish causation; we must perform multivariate analysis to ensure gender is not merely confounding with other risk factors like contract type or billing method.

Recommendation: Perform a segment slice of female customers by contract type (month-to-month) and internet service (fiber optic) to isolate and address the exact root cause of this high churn rate.

03 Rate By Subscription Type

What it shows: This horizontal bar chart displays the positive (churn) rate percentages across three subscription levels: Basic, Standard, and Premium.

Key finding: The churn rate is extremely high across all subscription tiers, ranging from a maximum of 48.3% for Basic customers to 46.5% for Premium customers, indicating a negligible variance of just 1.8 percentage points.

Business implication: Since churn rates remain high regardless of the subscription level, tier-specific benefits are not driving retention; the retention team should focus on broader service quality or pricing issues affecting all tiers.

Confidence caveat: This single-variable view does not account for confounding factors such as contract length or tenure, which are likely stronger drivers of churn.

Suggestion: Shift retention resources away from subscription tier incentives and focus instead on converting month-to-month contracts to multi-year commitments.

04 Rate By Contract Length

What it shows: This horizontal bar chart displays the positive (churn) rate percentage across three contract length categories: Monthly, Annual, and Quarterly.

Key finding: Monthly contracts exhibit the highest churn rate at 51.6%, which is 7.6 percentage points higher than Quarterly contracts (44.0%) and 5.4 percentage points higher than Annual contracts (46.2%).

Business implication: The retention team must prioritize campaigns to migrate high-risk Monthly subscribers to longer-term Quarterly or Annual contracts to stabilize recurring revenue.

Confidence caveat: This chart shows correlation rather than causation, meaning we need to analyze customer tenure and onboarding channels to confirm if contract length alone drives the churn difference.

Suggestion: Launch targeted incentive campaigns (e.g., discounts or feature upgrades) to transition Monthly users to Annual or Quarterly plans, while simultaneously auditing why baseline churn remains critically high (above 44%) across all contract types.

05 Distribution Age

What it shows: This histogram shows the density distribution of customer age, comparing retained customers (Negative, blue) and churned customers (Positive, orange), with dashed lines indicating their respective median ages of 41.0 and 44.0.

Key finding: Older customers have a higher propensity to churn, evidenced by a higher median age of 44.0 for the churned group compared to 41.0 for the retained group, with a distinct spike in churn density (above 0.030) among customers aged 51 to 65.

Business implication: Marketing and retention teams should launch tailored communication campaigns and simplified customer support channels aimed at retaining customers aged 50 and older.

Confidence caveat: This plot shows a simple bivariate correlation; further multivariate analysis is required to determine if age-related churn is driven by underlying factors like technology preferences or contract types.

Suggestion: Implement proactive outreach and personalized high-touch service for subscribers aged 50+ to mitigate their elevated churn risk.

06 Boxplot Age

What it shows: This boxplot displays the distribution of customer age across two target groups: Negative (retained) and Positive (churned). The y-axis shows customer age in years, while the x-axis visualizes the median, quartiles, and range for each group.

Key finding: The age distributions for both groups are highly similar, with churned customers (Positive) having a slightly higher median age of approximately 44 years compared to 41 years for retained customers (Negative). The interquartile ranges heavily overlap, spanning roughly 29–53 years for retained and 31–55 years for churned customers.

Business implication: Because the age profiles of churned and retained customers are nearly identical, age is not a viable primary metric for targeting retention campaigns. The retention team should focus efforts on more actionable operational variables, such as contract type or internet service.

Confidence caveat: While there is a slight upward shift in age for churned customers, the extensive overlap suggests this correlation is weak and requires a statistical t-test to confirm significance.

Suggestion: De-prioritize age-based segmentation in churn mitigation strategies and instead focus analytical resources on high-impact behavioral drivers like contract term and payment method.

07 Distribution Tenure

What it shows: This chart displays the density distribution of customer tenure in months, comparing the Negative target class (blue, median 27.0 months) against the Positive target class (orange, median 37.0 months) across a 60-month range.

Key finding: The Positive group has a significantly higher median tenure (37.0 months) than the Negative group (27.0 months), with a sharp, sustained increase in Positive class density occurring immediately after the 24-month threshold.

Business implication: Marketing and retention teams should design targeted loyalty interventions and contract-renewal incentives prior to the 24-month milestone to successfully transition customers into the higher-tenure bracket.

Confidence caveat: As this is a density plot, it visualizes relative proportions rather than absolute customer volumes; we need raw cohort sizes to calculate the exact revenue impact of these distributions.

Suggestion: Analyze this tenure distribution against the `contract` column to verify if the 24-month divergence is driven by the expiration of Two-year contracts, and target those specific contract holders for early renewal.

08 Boxplot Tenure

What it shows: This boxplot compares the distribution of customer tenure in months across two categories: "Negative" and "Positive" target groups. The y-axis measures tenure from 0 to 60 months, illustrating the median, quartiles, and range for each group.

Key finding: The "Positive" target group has a higher median tenure of approximately 37 months compared to 27 months for the "Negative" group. Furthermore, 75% of the "Positive" group has a tenure exceeding 25 months, while the "Negative" group's 25th percentile drops to 14 months.

Business implication: Assuming "Positive" represents retained customers, the retention team must focus campaigns on the critical 12-to-27-month window to prevent early-stage churn. Incentivizing customers to cross the 2-year milestone will significantly stabilize the customer base.

Confidence caveat: This chart shows a strong association between longer tenure and the positive class but does not control for confounding factors like contract types or promotional pricing.

Suggestion: Implement targeted loyalty interventions and contract-renewal incentives for customers at the 12-to-18-month tenure mark to successfully transition them into the lower-risk, long-term retention bracket.

09 Distribution Usage Frequency

What it shows: This density plot compares the distribution of monthly usage frequency for churned (Positive) versus retained (Negative) customers. The visual encodes usage frequency on the x-axis, density on the y-axis, and highlights their respective median values of 14.0 and 16.0.

Key finding: Churned customers have a significantly higher concentration at very low usage frequencies (1 to 5), reaching density levels over 0.06, resulting in a lower median usage frequency (14.0) than retained customers (16.0).

Business implication: The retention team must implement trigger-based re-engagement campaigns (e.g., feature tutorials or usage incentives) specifically targeting users whose activity drops below 6 times per month to arrest early-stage churn.

Confidence caveat: This chart shows correlation, not causation; additional qualitative feedback is required to determine if low usage is driven by onboarding friction or product dissatisfaction.

Suggestion: Implement automated, proactive email and in-app re-engagement playbooks specifically targeting customers with fewer than 6 usage sessions per month.

10 Boxplot Usage Frequency

What it shows: This boxplot compares the distribution of `usage_frequency` (scaled from 1 to 30) between the Negative (non-churned) and Positive (churned) customer groups.

Key finding: Churned customers (Positive) exhibit lower overall usage, with a median frequency of 14 compared to 16 for retained customers (Negative), and a significantly lower 25th percentile of 5 compared to 9.

Business implication: A drop in usage frequency is a leading indicator of churn; the customer success team should target accounts whose monthly usage drops below 5 with re-engagement campaigns.

Confidence caveat: This visualization displays a correlation but does not prove causation, and does not control for other factors like tenure or contract type.

Suggestion: Implement automated, proactive email triggers for customers whose usage frequency drops below 5 in a 30-day period to drive adoption and mitigate churn risk.

11 Distribution Support Calls

What it shows: This chart displays the density distribution of support calls (0 to 10) for retained (Negative, blue) versus churned (Positive, orange) customers, with vertical dashed lines marking their respective median values.

Key finding: Churned customers have a significantly higher median of 7.0 support calls compared to 4.0 for retained customers, with churn density spiking sharply to nearly 0.40 once support calls reach 5 or more.

Business implication: Exceeding 4 support calls is a critical churn trigger; the customer experience team must proactively intervene when a customer reaches this threshold to prevent further escalations.

Confidence caveat: This correlation does not establish direct causation, as we require data on call resolution success rates and specific issue categories to determine if unresolved problems are the primary driver of churn.

Suggestion: Implement an automated CRM alert system that flags accounts upon their 4th support call, routing them to a specialized retention team equipped with rapid-resolution tools and loyalty offers to mitigate churn risk before the 5-call tipping point.

12 Boxplot Support Calls

What it shows: This boxplot compares the distribution of support calls (y-axis) between two target groups (x-axis): Negative (retained) and Positive (churned) customers. The visualization encodes the median, quartiles, and range of support calls for each cohort.

Key finding: Customers who churned (Positive group) had a higher volume of support calls, with a median of 7 calls and 75% making 5 or more calls, compared to retained customers who had a median of 4 calls.

Business implication: High support call volume is a strong leading indicator of churn; the customer success team must proactively intervene and audit accounts once they reach 5 support calls to address underlying issues.

Confidence caveat: This represents correlation rather than causation; additional data on call resolution rates and customer sentiment is needed to verify if the call volume itself or

unresolved issues are driving the churn.

Suggestion: Implement an automated CRM alert that flags accounts and triggers proactive outreach by a specialized retention team once a customer reaches 5 support calls.

13 Distribution Payment Delay

What it shows: This density plot compares the distribution of payment delays (in days) between retained (Negative, blue) and churned (Positive, orange) customers. Dashed vertical lines represent the median payment delay for each group.

Key finding: Churned customers have a significantly higher median payment delay of 23.0 days compared to 12.0 days for retained customers, with a critical spike in churn density at the 30-day delay threshold.

Business implication: The retention and billing teams should implement targeted intervention strategies, such as automated dunning notifications or flexible payment options, before a payment delay reaches 15 days to prevent transition into high-risk churn territory.

Confidence caveat: While payment delays strongly correlate with churn, we need billing-method details to confirm if these delays are due to active customer neglect or passive issues like expired credit cards.

Suggestion: Implement proactive billing alerts at day 10 of payment delay and offer auto-pay incentives to customers utilizing manual payment methods to reduce friction.

14 Boxplot Payment Delay

What it shows: This boxplot compares the distribution of payment delay (in days) on the y-axis between two customer target groups on the x-axis: Negative (retained) and Positive (churned).

Key finding: Churned customers (Positive) have a significantly higher median payment delay of approximately 23 days, compared to a median delay of only 12 days for retained customers (Negative).

Business implication: Payment delay is a strong leading indicator of churn; the retention team should trigger automated payment reminders or proactive customer service outreach before delays cross the 18-day threshold (the 75th percentile for retained customers).

Confidence caveat: While payment delay correlates strongly with churn, this visual does not establish causation; delays could represent involuntary churn due to payment failure rather than voluntary cancellation.

Suggestion: Implement an automated, tiered notification workflow that triggers at 14 days of payment delay to offer flexible payment options and prevent involuntary churn.

15 Distribution Total Spend

What it shows: The chart displays the density distribution of total spend for retained

(Negative, blue) and churned (Positive, orange) customers, with dashed lines marking their respective median values.

Key finding: Churned customers have a lower median total spend (\$491.00) compared to retained customers (\$568.00), with a higher density of churn occurring below the \$500 spend threshold.

Business implication: The retention team should target accounts spending under \$500 with proactive value-add outreach or multi-service bundles to secure higher commitment and loyalty.

Confidence caveat: This association does not prove causation; lower total spend is highly likely to be a co-linear proxy for shorter customer tenure rather than an independent driver of churn.

Suggestion: Implement targeted onboarding campaigns for low-spend, month-to-month contracts to increase tenure beyond the critical early months.

16 Boxplot Total Spend

What it shows: This boxplot compares the distribution of total spend across the 'Negative' (retained) and 'Positive' (churned) target groups. The y-axis represents total spend from \$100 to \$1000, while the boxplots display the medians, quartiles, and ranges for each group.

Key finding: The median total spend for the 'Positive' (churned) group is lower at approximately \$490 compared to \$570 for the 'Negative' group, with both groups spanning the full range of \$100 to \$1000.

Business implication: Retention campaigns should focus on lower-spend cohorts, as they demonstrate a higher propensity to churn.

Confidence caveat: This bivariate analysis does not account for tenure; lower total spend in the churn group may simply reflect earlier churn rather than lower monthly budget.

Suggestion: Evaluate monthly charges instead of total spend to decouple customer value from tenure when predicting churn.

17 Correlation Heatmap

*Analysis failed: Gemini API error: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: <https://ai.google.dev/gemini-api/docs/rate-limits>. To monitor your current usage, head to: <https://ai.dev/rate-limit>.

- Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 20, model: gemini-3.5-flash Please retry in 56.206407176s.*
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18 Combo Gender Subscription Type

What it shows: This grouped bar chart displays the positive rate percentage across female and male demographics, segmented by three subscription types: Basic, Premium, and Standard.

Key finding: Females exhibit a significantly higher positive rate (averaging ~55%) compared to males (averaging ~38%), with Basic subscription holders showing the highest rates in both gender categories (56% for females, 39% for males).

Business implication: Retention teams must prioritize female subscribers, specifically launching proactive engagement campaigns for those on Basic plans to address this disproportionate risk.

Confidence caveat: This chart shows correlation only; multi-variable regression is required to control for tenure and monthly charges to ensure gender is the true driver of this variance.

Suggestion: Implement a targeted retention pilot offering subscription-tier upgrades or loyalty incentives specifically designed for female customers on Basic and Standard plans to close the 17% gender gap.